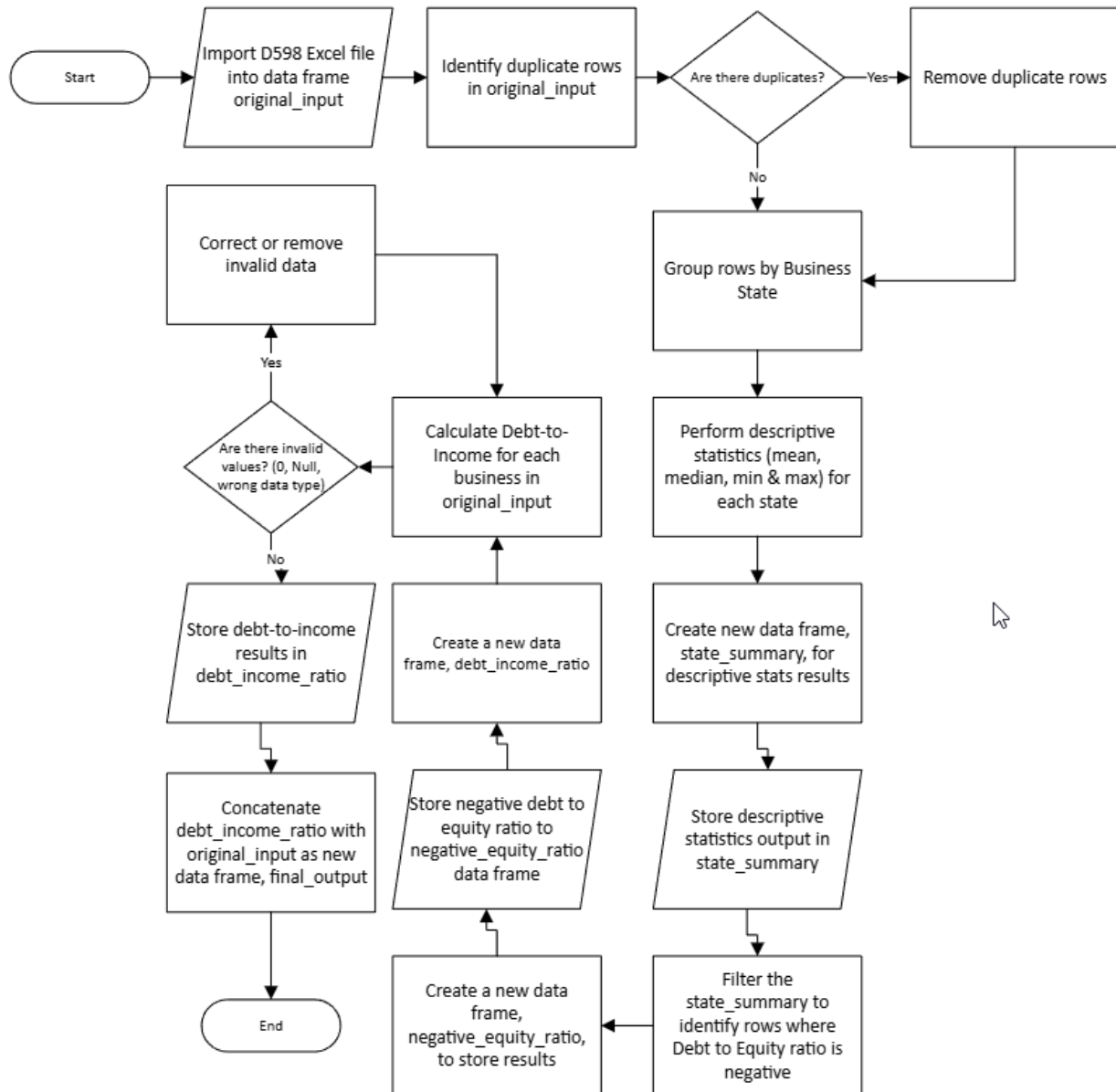


D598 Task 1: Program Planning

A. Create a flowchart for a program to perform the required task.



B. Write pseudocode for a program to perform the required task.

Note: The provided Excel file is named D598 Data Set and includes the following columns:

Business ID, Business State, Total Long-term Debt, Total Equity, Debt to Equity, Total Liabilities, Total Revenue, and Profit Margin.

Start

Import the Excel D598 Data Set into a new data frame, `original_input`

duplicate handling

Identify duplicate rows in `original_input`

Remove duplicates from the data frame

#descriptive stats by state

Group data in `original_input` by Business State

For each state, calculate mean, median, minimum, and maximum for:

- Total Long-Term Debt
- Total Equity
- Debt to Equity
- Total Liabilities
- Total Revenue
- Profit Margin

Store the results in a new data frame, `state_summary`

#filter for negative debt-to-equity

Filter `state_summary` to show only rows where Debt to Equity ratio is negative

Store the results in a new data frame, `negative_equity_ratio`

```

#calculate debt-to-income

Create a new data frame, debt_income_ratio, as a copy of original_input

For each business in debt_income_ratio:

    Calculate the debt-to-income ratio = Total Long-term Debt / Total Revenue

#data validation

    Check original_input for invalid values (e.g., 0, Null, wrong data type)

    Correct or remove invalid data

    Store results as a new data frame, debt_income_ratio

#concatenate results

    Concatenate the data frame debt_income_ratio with original_input to create a new data frame,
final_output

    END

```

C. Provide an explanation of the relationship between the flowchart and pseudocode that does the following:

1. Describe the logic behind the flowchart and pseudocode.

Flowcharts and pseudocode are essential tools for planning the structure and logic of a program before implementation. In this case, the flowchart and pseudocode represent the same sequence of actions required to prepare and analyze the provided D598 Data Set from an Excel file.

The flowchart provides a high-level visual breakdown of the process logic. Starting with importing the dataset, removing duplicate rows, grouping data by business state, and performing descriptive statistics. Decision points such as checking for invalid data are included and branch accordingly to correct or skip as needed. After validating and cleaning the data, the flow calculates new metrics like debt-to-income

ratio and filters rows with negative debt-to-equity values. Finally, it shows how the program will combine the processed data into a final output for later analysis.

The pseudocode mirrors this logic in a text format. It breaks down the same steps line by line. Where the flowchart uses visual decision diamonds and arrows to represent logic flow, the pseudocode uses conditional statements and sequential commands.

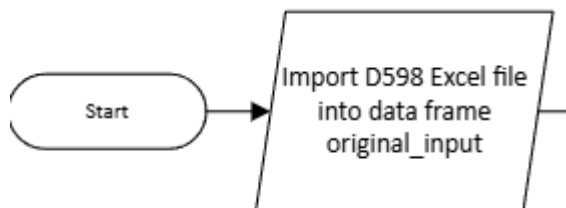
2. Explain the alignment between flowchart and pseudocode.

Flowcharts and pseudocode are complementary tools in planning algorithms and aid in turning complex problems into more manageable steps. The flowchart provides a high-level visual representation of the algorithm and its decision points. Pseudocode provides a more detailed text-based breakdown of the steps the algorithm will follow and the processes it must complete.

Comparing the above flowcharts and pseudocode, we can see how they align.

Start

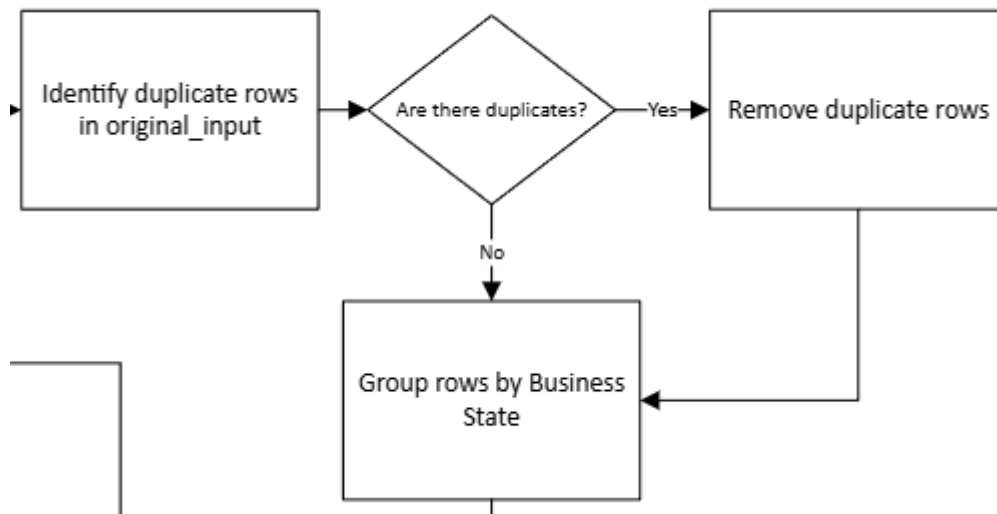
Import the Excel D598 Data Set into a new data frame, original_input



duplicate handling

Identify duplicate rows in original_input

Remove duplicates from the data frame



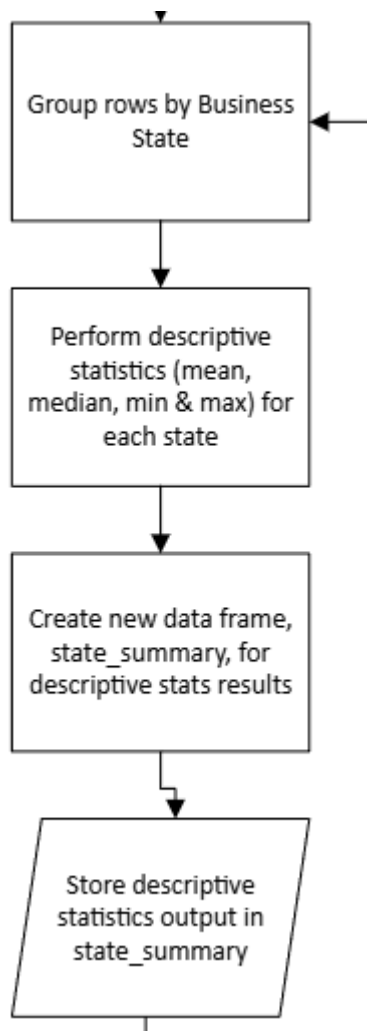
#descriptive stats by state

Group data in original_input by Business State

For each state, calculate mean, median, minimum, and maximum for:

- Total Long-Term Debt
- Total Equity
- Debt to Equity
- Total Liabilities
- Total Revenue
- Profit Margin

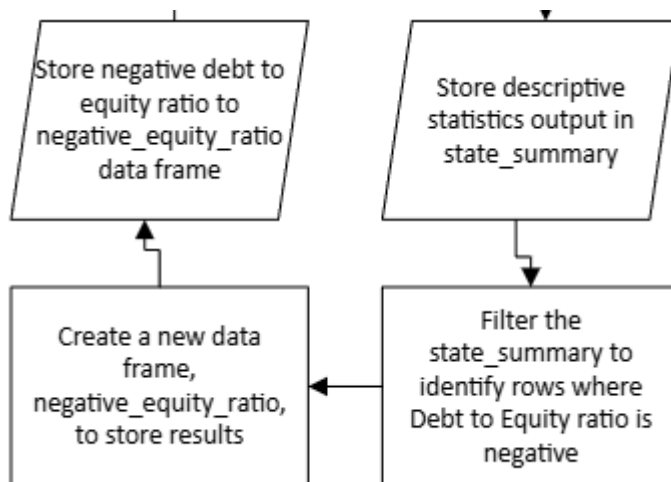
Store the results in a new data frame, state_summary



#filter for negative debt-to-equity

Filter state_summary to show only rows where Debt to Equity ratio is negative

Store the results in a new data frame, negative_equity_ratio



#calculate debt-to-income

Create a new data frame, debt_income_ratio, as a copy of original_input

For each business in debt_income_ratio:

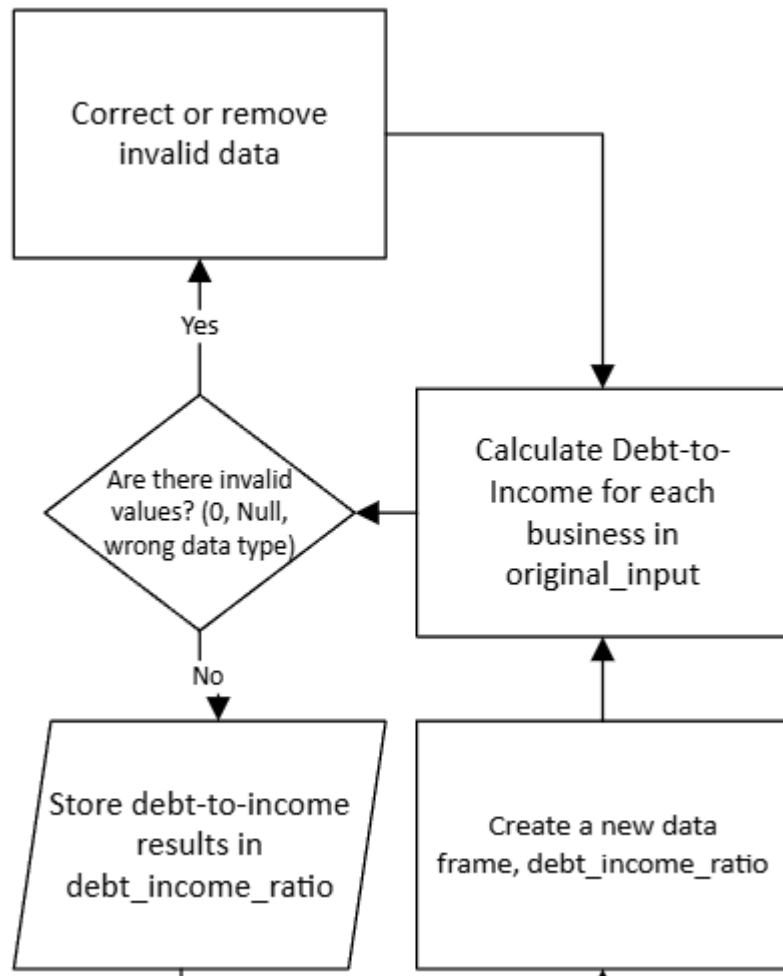
Calculate the debt-to-income ratio = Total Long-term Debt / Total Revenue

#data validation

Check original_input for invalid values (e.g., 0, Null, wrong data type)

Correct or remove invalid data

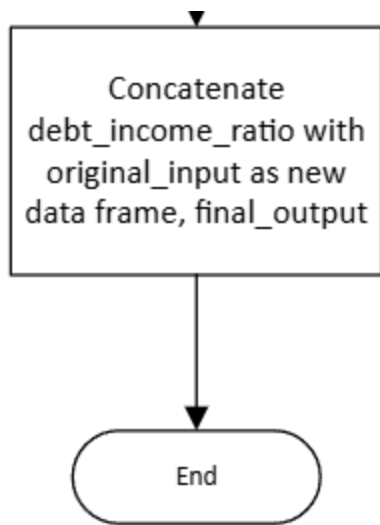
Store results as a new data frame, debt_income_ratio



#concatenate results

Concatenate the data frame `debt_income_ratio` with `original_input` to create a new data frame,
`final_output`

END



References

Pseudocode and flowchart: Complete Beginner's Guide. Codecademy. (n.d.).

<https://www.codecademy.com/article/pseudocode-and-flowchart-complete-beginners-guide>

WGU Analytics Programming course materials, July 2025